

CyberPower Hybrid Inverter Modbus Protocol

Support for
CPSHB6000ETL48
CPSHB6000ETLFR

Revision History

Rev.	Date	Description of Change	Protocol Version	Fault Table Version	Author
V1.00	2018/04/17	First draft	1.00	1.00	Jim.Kuo
V1.01	2018/10/19	Add Fault Table description	1.01	1.00	Jim.Kuo
V1.02	2019/12/11	Add String's E-today & E-total (0x2070~0x209F)	1.01	1.00	Jim.Kuo
V1.03	2021/12/13	Fix Some Bug	1.01	1.00	Jim.Kuo
V1.04	2022/06/10	Fix Some Bug	1.01	1.00	Jim.Kuo

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1. Introduction

This document explains the implementation of the Modbus communications protocol specific to CyberPower Systems Hybrid inverters. The Modbus protocol is an industry standard that allows a master station to interrogate devices for its publically available data and set supported control and configuration parameters.

2. Related documents

<http://www.modbus.org/> Modicon Modbus Protocol Reference Guide

<http://www.modbus.org/> Modbus over Serial Line Specification and Implementation Guide, V1.2

<http://www.modbus.org/> Modbus Application Protocol Specification, V1.1b

<http://www.sunspec.org/> SunSpec Alliance Specifications

3. Modbus Configuration

3.1. Supported Transmission Medium

Products supporting Modbus communicate primarily via the RS-485 (TIA/EIA-485-A) communications standard. The RS-485 medium allows multiple devices to communicate on a single medium using a master/slave approach for arbitrating the bus. All CyberPower Systems products are 1/8 unit load (UL) devices. Depending on the mix of other devices from other vendors, biasing or polarization of devices on the network, this will allow for as few as 32 devices or as many as 255 devices on a single bus.

3.2. Supported Modes of Transmission

The Modbus protocol supports two serial transmission modes, ASCII and RTU (Remote Terminal Unit). CyberPower Systems only supports the RTU mode of operation.

3.3. Communications Parameters

Serial communications default to 8 data bits, no parity, and 1 stop bit. A default baud rate of 9600 BPS is used. Parameters are reconfigurable.

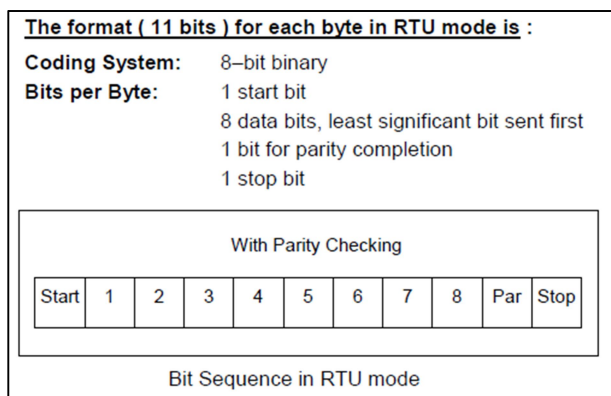


Figure 3-1 The format for each byte in RTU mode

3.4. Exception Responses

Except for broadcast messages, when a master device sends a query to a slave device it expects a normal response. One of four possible events can occur from the master's query:

- If the slave device receives the query without a communication error, and can handle the query normally, it does not return any response.
- If the slave does not receive the query due to a communication error, no response is returned. The master program will eventually process a timeout condition for the query.
- If the slave receives the query, but detects a communication error (parity, LRC, or CRC), no response is returned. The master program will eventually process a timeout condition for the query.
- If the slave receives the query without a communication error, but cannot handle it (for example, if the request

is to read a non-existent coil or register), the slave will return an exception response informing the master of the nature of the error.

The table below describes the exception codes supported by CyberPower Systems devices along with their possible causes.

Code	Name	Meaning
01	ILLEGAL FUNCTION	The function code received in the query is not an allowable action for the slave. If a Poll Program Complete command was issued, this code indicates that no program function preceded it.
02	ILLEGAL DATA ADDRESS	The data address received in the query is not an allowable address for the slave.
03	ILLEGAL DATA VALUE	A value contained in the query data field is not an allowable value for the slave.
04	SLAVE DEVICE FAILURE	An unrecoverable error occurred while the slave was attempting to perform the requested action.
06	SLAVE DEVICE BUSY	The slave is engaged in processing a long-duration program command. The master should retransmit the message later when the slave is free.

Table 3-1 Exception Codes Supported by CyberPower Systems Devices

3.5. CRC Generation

When RTU mode is used for character framing, the error checking field contains a 16-bit value implemented as two 8-bit bytes. The error check value is the result of a Cyclical Redundancy Check calculation performed on the message contents.

The CRC field is appended to the message as the last field in the message. When this is done, the low-order byte of the field is appended first, followed by the high-order byte. The CRC high-order byte is the last byte to be sent in the message.

The CRC value is calculated by the transmitting device, which appends the CRC to the message. The receiving device recalculates a CRC during receipt of the message, and compares the calculated value to the actual value it received in the CRC field. If the two values are not equal, an error results.

3.6. Addressing Rules

The Modbus addressing space comprises 256 different addresses. The addresses from 248 to 255 are reserved and the address 0 is broadcast address.

All CyberPower Systems products support broadcast request packets from the master. As its name implies, broadcasts allow all devices to receive and process the same command from the Modbus master. Broadcasts are triggered by setting the slave address to 0 (0x00).

0	From 1 to 247	From 248 to 255
Broadcast address	Slave individual addresses	Reserved

Table 3-2 Addressing rules

3.7. RTU Framing

In RTU mode, messages start with a silent interval of at least 3.5 character times. This is most easily implemented as a multiple of character times at the baud rate that is being used on the network. The first field then transmitted is the device address.

The allowable characters transmitted for all fields are hexadecimal 0–9, A–F. Networked devices monitor the network bus continuously, including during the ‘silent’ intervals. When the first field (the address field) is received, each device decodes it to find out if it is the addressed device. Following the last transmitted character, a similar interval of at least 3.5 character times marks the end of the message. A new message can begin after this interval.

The entire message frame must be transmitted as a continuous stream. If a silent interval of more than 1.5 character times occurs before completion of the frame, the receiving device flushes the incomplete message and assumes that the next byte will be the address field of a new message. Similarly, if a new message begins earlier than 3.5 character times following a previous message, the receiving device will consider it a continuation of the previous message. This will set an error, as the value in the final CRC field will not be valid for the combined messages.

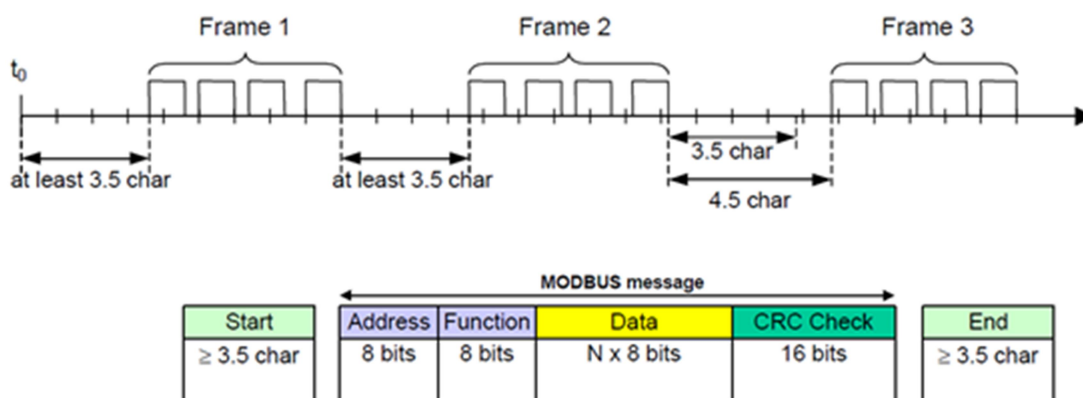


Figure 3-2 RTU Message Frame (1)

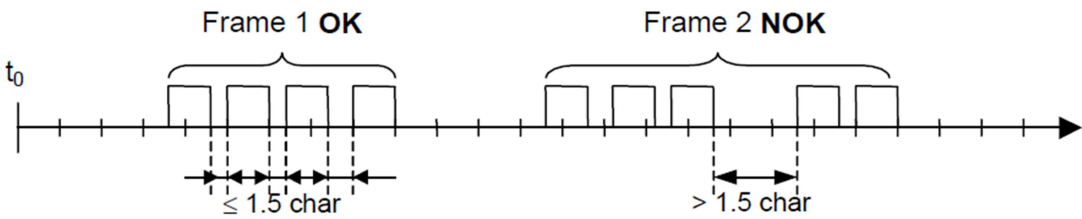


Figure 3-3 RTU Message Frame (2)

4. Supported Modbus Functions

4.1. Function 1 (0x01): Read Coils

Description

Reads the ON/OFF status of discrete outputs.

Query

The query message specifies the starting coil and quantity of coils to be read.

Response

The coil status in the response message is packed as one coil per bit of the data field. Status is indicated as: 1 = ON; 0 = OFF. The LSB of the first data byte contains the coil addressed in the query. The other coils follow toward the high order end of this byte, and from 'low order to high order' in subsequent bytes.

If the returned coil quantity is not a multiple of eight, the remaining bits in the final data byte will be padded with zeros (toward the high order end of the byte). The Byte Count field specifies the quantity of complete bytes of data.

Here is an example of a request to read Coils 6000h – 600Fh:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	01	Function	01
Starting Address Hi	60	Byte Count	02
Starting Address Lo	00	Outputs status 6000-6007	00
Quantity of Outputs Hi	00	Outputs status 6008-600F	00
Quantity of Outputs Lo	10	CRC Lo	B9
CRC Lo	23	CRC Hi	FC
CRC Hi	C6		

Table 4-1 example of a request to read Coils

4.2. Function 2 (0x02): Read Discrete Inputs

Description

Reads the ON/OFF status of discrete inputs (1X references) in the slave.

Query

The query message specifies the starting input and quantity of inputs to be read.

Response

The input status in the response message is packed as one input per bit of the data field. Status is indicated as: 1 = ON; 0 = OFF. The LSB of the first data byte contains the input addressed in the query. The other inputs follow toward the high order end of this byte, and from 'low order to high order' in subsequent bytes.

If the returned input quantity is not a multiple of eight, the remaining bits in the final data byte will be padded with zeros (toward the high order end of the byte). The Byte Count field specifies the quantity of complete bytes of data.

Here is an example of a request to read Discrete Inputs 5000h – 500Fh:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	02	Function	02
Starting Address Hi	50	Byte Count	02
Starting Address Lo	00	Outputs status 5000-5007	00
Quantity of Outputs Hi	00	Outputs status 5008-500F	00
Quantity of Outputs Lo	10	CRC Lo	B9
CRC Lo	68	CRC Hi	B8
CRC Hi	C6		

Table 4-2 example of a request to read Discrete Inputs

4.3. Function 3 (0x03): Read Holding Registers

Description

Reads the binary contents of holding registers (4X references) in the slave.

Query

The query message specifies the starting register and quantity of registers to be read.

Response

The register data in the response message are packed as two bytes per register, with the binary contents right justified within each byte. For each register, the first byte contains the high order bits and the second contains the low order bits.

Here is an example of a request to read Holding Registers 0532h – 0533h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>

Slave ID	01	Slave ID	01
Function	03	Function	03
Starting Address Hi	05	Byte Count	04
Starting Address Lo	32	Register value Hi (0532)	07
No. of Registers Hi	00	Register value Lo (0532)	D0
No. of Registers Lo	02	Register value Hi (0533)	00
CRC Lo	65	Register value Lo (0533)	01
CRC Hi	08	CRC Lo	3B
		CRC Hi	7E

Table 4-3 example of a request to read Holding Registers

4.4. Function 4 (0x04): Read Input Registers

Description

Reads the binary contents of input registers (3X references) in the slave.

Query

The query message specifies the starting register and quantity of registers to be read.

Response

The register data in the response message are packed as two bytes per register, with the binary contents right justified within each byte. For each register, the first byte contains the high order bits and the second contains the low order bits.

Here is an example of a request to read Input Registers 1000h – 1001h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	04	Function	04
Starting Address Hi	10	Byte Count	04
Starting Address Lo	00	Register value Hi (0532)	CC
No. of Registers Hi	00	Register value Lo (0532)	94
No. of Registers Lo	02	Register value Hi (0533)	06
CRC Lo	75	Register value Lo (0533)	0C
CRC Hi	0B	CRC Lo	86
		CRC Hi	9D

Table 4-4 example of a request to read Input Registers

4.5. Function 5 (0x05): Write Single Coil

Description

Forces a single coil to either ON or OFF. When broadcast, the function forces the same coil reference in all attached slaves.

Query

The query message specifies the coil reference to be forced. The requested ON/OFF state is specified by a constant in the query data field. A value of FF 00 hex requests the coil to be ON. A value of 00 00 requests it to be OFF. All other values are illegal and will not affect the coil.

Response

The normal response is an echo of the query, returned after the coil state has been forced. Here is an example of a request to write Coil 6000h ON:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	05	Function	05
Output Address Hi	60	Output Address Hi	60
Output Address Lo	00	Output Address Lo	00
Output Value Hi	FF	Output Value Hi	FF
Output Value Lo	00	Output Value Lo	00
CRC Lo	92	CRC Lo	92
CRC Hi	3A	CRC Hi	3A

Table 4-5 example of a request to write single Coil

4.6. Function 6 (0x06): Write Single Register

Description

This function code is used to write a single holding register in a remote device. When broadcast, the function presets the same register reference in all attached slaves.

Query

The query message specifies the register reference to be preset. The requested preset value is specified in the query data field.

Response

The normal response is an echo of the request, returned after the register contents have been written.

Here is an example of a request to write Holding Register 0512h to 03E8h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	06	Function	06
Register Address Hi	05	Register Address Hi	05
Register Address Lo	12	Register Address Lo	12
Register Value Hi	03	Register Value Hi	03
Register Value Lo	E8	Register Value Lo	E8
CRC Lo	29	CRC Lo	29
CRC Hi	BD	CRC Hi	BD

Table 4-6 example of a request to write Holding Register

4.7. Function 16 (0x10): Write Multiple Registers

Description

This function code is used to write a block of contiguous registers (1 to 123 registers). When broadcast, the function presets the same register references in all attached slaves.

Query

The requested written values are specified in the request data field. Data is packed as two bytes per register.

Response

The normal response returns the function code, starting address, and quantity of registers written.

Here is an example of a request to write three registers at 0532h to 07DCh, 0008h and 0014h:

Request		Response	
<i>Field Name</i>	<i>(Hex)</i>	<i>Field Name</i>	<i>(Hex)</i>
Slave ID	01	Slave ID	01
Function	10	Function	10
Starting Address Hi	05	Starting Address Hi	05
Starting Address Lo	32	Starting Address Lo	32
Quantity of Registers Hi	00	Quantity of Registers Hi	00
Quantity of Registers Lo	03	Quantity of Registers Lo	03
Byte Count	06	CRC Lo	21
Registers Value Hi	07	CRC Hi	0B
Registers Value Lo	DC		
Registers Value Hi	00		
Registers Value Lo	08		

Registers Value Hi	00		
Registers Value Lo	14		
CRC Lo	07		
CRC Hi	D0		

Table 4-7 example of a request to write three registers

5. Modbus Register Map

5.1. Modbus Data Formats

The MODBUS specification is not explicit on how to encode numbers other than 16 bit integers. Differences do exist between one manufacturer's implementation and another's. **This specification restricts values to the following standard data formats and it is difference to the formats in "SunSpec MODBUS Register Mappings".**

16 bit Integer Values

Values are stored in big-endian order per the MODBUS specification and consist of a single register. All integer values are documented a signed or unsigned. All signed values are represented using two's-compliment.

Modbus Register	1															
Byte	0								1							
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

32 bit Integer Values

32 bit integers are stored using two registers in big-endian order

Modbus Register	1								2							
Byte	0				1				2				3			
Bits	15 ... 8				7 ... 0				31 ... 24				23 ... 16			

64 bit Integer Values

64 bit integers are stored using four registers in big-endian order.

Modbus Register	1								2							
Byte	0				1				2				3			
Bits	15 ... 8				7 ... 0				31 ... 24				23 ... 16			

Modbus Register	3								4							
Byte	4				5				6				7			
Bits	47 ... 40				39 ... 32				63 ... 56				55 ... 48			

String Values

Store variable length string values in a fixed size register range using a NULL (0 value) to terminate or pad the string. For example, up to 16 characters can be stored in 8 contiguous

registers as follows
EX : "EXAMPLE STRING!"

Modbus Register	1		2		3		4		5		6		7		8	
Byte	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Character	E	X	A	M	P	L	E		S	T	R	I	N	G	!	NULL

NOT_IMPLEMENTED value: all registers filled with NULL or 0x0000.

Floating Point Values,

Floating point values are 32 bits and encoded according to the IEEE 754 floating point standard.

Modbus Register	1		2		
Byte	0	1	2	3	
Bits	15 ... 8	7 ... 0	31	30 ... 24	23 ... 16
IEEE 754	Fraction least		sign	Exponent	Fraction

5.2. Input Registers (0x04)

5.2.1. Monitoring Parameters

Address (Hex)	Name	Type	Unit	Remark
2100 ~2107	Reserved			
2108 ~210F	Serial Number	uint16		
2110 ~2117	Model Name	uint16		
2118 ~2135	Reserved			
2136 2137	FeedInGrid Life energy	uint32	0.1kWh	
2138 2139	Inv. OP today energy	uint32	0.1kWh	
213A 213B	Reserved			
213C 213D	Inv. OP life energy	uint32	0.1kWh	
213E 213F	Reserved			
2140 2141	PV1 today energy	uint32	0.1kWh	
2142 2143	PV1 life energy	uint32	0.1kWh	
2144 2145	PV2 today energy	uint32	0.1kWh	
2146 2147	PV2 life energy	uint32	0.1kWh	
2148 2149	PV3 today energy	uint32	0.1kWh	
214A 214B	PV3 life energy	uint32	0.1kWh	
214C 214D	PV4 today energy	uint32	0.1kWh	

214E	PV4 life energy	uint32	0.1kWh	
214F				
2150	Feed In Grid today energy	uint32	0.1kWh	
2151				
2152	Grid to Load today energy	uint32	0.1kWh	
2153				
2154	Grid to Load life energy	uint32	0.1kWh	
2155				
2156	EPS today energy	uint32	0.1kWh	
2157				
2158	EPS life energy	uint32	0.1kWh	
2159				
215A	Reserved	uint32	0.1kWh	
215B				
215C	Battery charge today energy	uint32	0.1kWh	
215D				
215E	Battery discharge today energy	uint32	0.1kWh	
215F				
2160	Veps_A	uint16	0.1V	
2161	Reserved			
2162	Reserved			
2163	Ieps_A	uint16	0.01A	
2164	Reserved			
2165	Reserved			
2166	Peps_A	uint16	1W	
2167	Reserved			
2168	Reserved			
2169	VAeps_A	uint16	1VA	
216A	Reserved			
216B	Reserved			
216C	Reserved			
216D	Reserved			
216E	Reserved			
216F	Feps_A	uint16	0.01Hz	
2170	Reserved			
2171	Reserved			
2172	Reserved			
2173	Reserved			
2174	Reserved			

2175				
2176	Battery Charge life energy	uint32	0.1kWh	
2177				
2178	Battery discharge life energy	uint32	0.1kWh	
2179				
217A	Vbat	uint16	0.1V	
217B	Reserved			
217C	Reserved			
217D	Ibat	int16	0.01A	Negative means discharge
217E	Pbat	int16	1W	
217F	Bat SOC	uint16	1%	
2180	Bat SOH	Uint16	1%	
2181	Reserved			
2182	Vgrid_A	uint16	0.1V	
2183	Reserved			
2184	Reserved			
2185	Igrid_A	uint16	0.01A	
2186	Reserved			
2187	Reserved			
2188	Pgrid_A	Int32	1W	Negative means feed in grid
2189				
218A	Reserved			
218B				
218C	Reserved			
218D				
218E	Reserved			
218F				
2190	Reserved			
2191				
2192	Reserved			
2193				
2194	PV1 Voltage	uint16	0.1V	
2195	PV1 Current	uint16	0.01A	
2196	PV1 Power	uint16	1W	
2197	PV2 Voltage	uint16	0.1V	
2198	PV2 Current	uint16	0.01A	
2199	PV2 Power	uint16	1W	
219A	PV3 Voltage	uint16	0.1V	
219B	PV3 Current	uint16	0.01A	

219C	PV3 Power	uint16	1W	
219D	PV4 Voltage	uint16	0.1V	
219E	PV4 Current	uint16	0.01A	
219F	PV4 Power	uint16	1W	
21A0	Inv phase-A voltage	uint16	0.1V	
21A1	Inv phase-B voltage	uint16	0.1V	
21A2	Inv phase-C voltage	uint16	0.1V	
21A3	Inv phase-A output current	uint16	0.01A	
21A4	Inv phase-B output current	uint16	0.01A	
21A5	Inv phase-C output current	uint16	0.01A	
21A6	Inv phase-A output power	uint16	1W	
21A7	Inv phase-B output power	uint16	1W	
21A8	Inv phase-C output power	uint16	1W	
21A9	Inv phase-AB voltage	uint16	0.1V	
21AA	Inv phase-BC voltage	uint16	0.1V	
21AB	Inv phase-CA voltage	uint16	0.1V	
21AC	Grid Frequency	uint16	0.01Hz	
21AD	Reserved			
21AE	Reserved			
21AF	Reserved			
21B0	Reserved			
21B2	Pload	uint16	1W	
21B3				
21B4	Load today energy	uint32	0.1kWh	
21B5				
21B6	Load life energy	uint32	0.1kWh	
21B7				
21B8	Reserved			
21B9	BUS1_Voltage	uint16	0.1V	
21BA	BUS2_Voltage	uint16	0.1V	

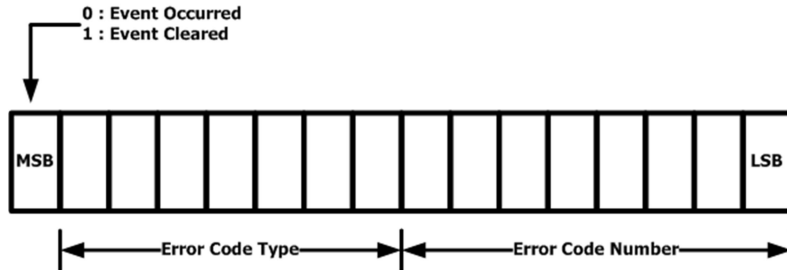
21BB	PV Rleakage	uint16	1kohm	
21BC	Temperature (air inside around power stage)	int16	0.1°C	
21BD	Temperature (air inside of wiring box)	int16	0.1°C	
21BE	Highest inverter temperature	int16	0.1°C	
21BF	Highest D/D temperature	int16	0.1°C	
21C0	Reserved	uint16		
21C1	Reconnection Time	uint16	1sec	
21C2	I_Event_flags1	uint32	bitfield	Event flags for Sunspec protocol
21C3				
21C4	I_Event_flags2	uint32	bitfield	Not supported
21C5				
21C6	I_Event_1_Vendor	uint16	bitfield	Event Flags (E00 ~ E15)
21C7	I_Event_2_Vendor	uint16	bitfield	Event Flags (E16 ~ E31)
21C8	I_Event_3_Vendor	uint16	bitfield	Event Flags (W00 ~ W15)
21C9	I_Event_4_Vendor	uint16	bitfield	Event Flags (F00 ~ F15)
21CA	I_Event_5_Vendor	uint16	bitfield	Event Flags (F16 ~ F31)
21CB	I_Event_6_Vendor	uint16	bitfield	Event Flags (F32 ~ F47)
21CC	I_Event_7_Vendor	uint16	bitfield	Event Flags (F48 ~ F63)
21CD	I_Event_8_Vendor	uint16	bitfield	Event Flags (F64 ~ F79)
21CE	I_Event_9_Vendor	uint16	bitfield	Event Flags (W16 ~ W31)
21CF ~21DF	Reserved			
21D0	Vbms	uint16		
21D1	Ibms	int16		
21D2	Pbms	int16		
21D3	Tbms	int16		
21D4	Bms charge Volt reference	uint16		
21D5	Bms charge Curr reference	uint16		
21D6	Bms discharge Curr reference	uint16		
21D7	Bms charge status	uint16		
21D8	Bms Warning_1	uint16		

[illegible]

[illegible]

Appendix A: Error Code

1. Error Code Format



Error Code Type	
E code: Error	45h
W code : Warning	57h
F code : Fault	46h

For example:

- "AC Over Frequency" exist: E07 => 4507h

2. "E" Error Code Table

Error Code	Hex Code	Event	Remark
E00	4500	No Grid	No Utility
E01	4501	Grid over frequency	AC Frequency above limit
E02	4502	Grid under frequency	AC Frequency under limit
E03	4503	Phase-A over voltage	AC Phase A Voltage above limit
E04	4504	Phase-A under voltage	AC Phase A Voltage under limit
E05	4505	Reserved	
E06	4506	Reserved	
E07	4507	Reserved	
E08	4508	Reserved	
E09	4509	Reserved	
E10	450A	Reserved	
E11	450B	Reserved	
E12	450C	Reserved	
E13	450D	Reserved	
E14	450E	Reserved	
E15	450F	Reserved	
E16	4510	Reserved	
E17	4511	Reserved	
E18	4512	Reserved	
E19	4513	Bat Reversed	Battery reversed
E20	4514	PV1 over voltage	Input1 DC Voltage above limit
E21	4515	PV2 over voltage	Input2 DC Voltage above limit

E22	4516	PV3 over voltage	Input3 DC Voltage above limit
E23	4517	PV4 over voltage	Input4 DC Voltage above limit
E24	4518	Par. Phase Err	Eps Phase error in parral mode
E25	4519	Reserved	
E26	451A	Reserved	
E27	451B	Reserved	
E28	451C	Reserved	
E29	451D	Reserved	
E30	451E	Reserved	
E31	451F	Reserved	

3. "W" Error Code Table

Error Code	Hex Code	Event	Remark
W00	5700	PV1 under voltage	Input1 DC Voltage under limit
W01	5701	PV2 under voltage	Input2 DC Voltage under limit
W02	5702	PV3 under voltage	Input3 DC Voltage under limit
W03	5703	PV4 under voltage	Input4 DC Voltage under limit
W04	5704	Reserved	
W05	5705	Reserved	
W06	5706	Reserved	
W07	5707	Reserved	
W08	5708	Power de-rating	Reduce output power due to temperature high or else
W09	5709	Reserved	
W10	570A	Fan1 Locked	Fan Failed
W11	570B	Reserved	
W12	570C	EPS Overload	
W13	570D	CT Reversed	
W14	570E	No Load or CT Disconnected	
W15	570F	Bat Noconnect	
W16	5710	Reserved	
W17	5711	Bat SOC Low	Battery's SOC under limit
W18	5712	Bat SOH Low	Battery's SOH under limit
W19	5713	Fan2 Locked	
W20	5714	BMS Comm Fail	

W21	5715	BMS Warnings	
W22	5716	AC Overload	Peps above limit
W23	5717	Reserved	
W24	5718	Reserved	
W25	5719	Reserved	
W26	571A	Reserved	
W27	571B	Reserved	
W28	571C	Reserved	
W29	571D	Reserved	
W30	571E	Reserved	
W31	571F	Reserved	

4. "F" Error Code Table

Error Code	Hex Code	Event	Remark
F00	4600	Reserved	
F01	4601	Grid output DC over current	AC Current DC Component above limit
F02	4602	Reserved	
F03	4603	Reserved	
F04	4604	Reserved	
F05	4605	Temperature low	Temperature under limit
F06	4606	Temperature high	Temperature above limit
F07	4607	Temperature sensor fail	Temperature sensor error, or heating circuit failure
F08	4608	Reserved	
F09	4609	Reserved	
F10	460A	Reserved	
F11	460B	Reserved	
F12	460C	Reserved	
F13	460D	Reserved	
F14	460E	Firmwares are not compatible	The firmware are not compatiabile
F15	460F	DSP ADC1	DSP ADC1 sensor error or circuit failure
F16	4610	DSP ADC2	DSP ADC2 sensor error or circuit failure
F17	4611	DSP ADC3	DSP ADC3 sensor error or circuit failure
F18	4612	DSP ADC4	DSP ADC4 sensor error or circuit failure
F19	4613	Red. ADC1	Redundant MCU ADC1 sensor error or circuit failure
F20	4614	Efficiency	HW Efficiency fail
F21	4615	Reserved	

F22	4616	HW Com1	The serial communication port between DSP and comm. MCU is failure
F23	4617	HW Com2	The serial communication port between DSP and redundant MCU is failure
F24	4618	Ground current	Residual current is above limit
F25	4619	RCMU fail	Residual current sensor error or circuit failure
F26	461A	Insulation fail	Insulation resistance to Earth is under limit
F27	461B	Reserved	
F28	461C	Relay short	Relay or drive circuit failure (Relay Short)
F29	461D	Relay open	Relay or drive circuit failure (Relay Open)
F30	461E	Bus unbalanced	Bus1 and Bus2 voltage are not balance
F31	461F	Bus+ OVR	Bus1 voltage above limit or circuit failure
F32	4620	Bus+ UVR	Bus1 voltage under limit or circuit failure
F33	4621	Bus- OVR	Bus2 voltage above limit or circuit failure
F34	4622	Bus- UVR	Bus2 voltage under limit or circuit failure
F35	4623	Total Bus OVR	Bus voltage above limit or circuit failure
F36	4624	Reserved	
F37	4625	Reserved	
F38	4626	Reserved	
F39	4627	Reserved	
F40	4628	Reserved	
F41	4629	Reserved	
F42	462A	CT fail	Inverter Current sensor or circuit failure
F43	462B	Reserved	
F44	462C	Reserved	
F45	462D	AC HW OCP	Hardware Inverter Curentt over current protection
F46	462E	Aux PWR Fail	Aux PWR Fail
F47	462F	HW D/D Fail	HW D/D Fail
F48	4630	Manual shutdown	Manual shutdown
F49	4631	Wiring fault	
F50	4632	CAN Set Fail	
F51	4633	EPS Abnormal	
F52	4634	CAN Fail	
F53	4635	Bat OVR	
F54	4636	Bat UVR	
F55	4637	EPS Overload	Peps above limit
F56	4638	EPS OP Short	
F57	4639	EPS OP OVR	
F58	463A	EPS OP UVR	

F59	463B	Par. Phase Loss	Eps loss phase in parral mode
F60	463C	PV Input over current	Input Current above limit
F61	463D	Reserved	
F62	463E	Reserved	
F63	463F	Reserved	
F64	4640	BMS Falut	
F65	4641	Reserved	
F66	4642	Reserved	
F67	4643	Reserved	
F68	4644	Reserved	
F69	4645	Other	Not defined fault
F70	4646	Reserved	
F71	4647	Reserved	

Appendix B: Product ID

Product ID

Code	Definition	Grid-tied PV Inverter (5K)	Grid-tied PV Inverter (TL7K-20K)	Grid-tied PV Inverter (TL50K)	CPSHB6000ETL48
0	Modbus Protocol version, 0: old version 1: new version	0	0	1	1
1	Product type, 1: Grid-tied 2: Hybrid 3: Off-grid 4: Solar charger	1	1	1	2
2	Isolation (TR/TL), 0: TL 1: TR	0	0	0	0
3	Output type, 0: Three phase 1: Single phase 2: DC output	1	0	0	1
4-6	Power, AAB: AA x 10^B	502	Depends	503	Depends
7	1: CPS 2: CPS(Co-operate with Growatt)	1	2	1	1
8	0: Default	0	0	0	0
9	reserved(0)	0	0	0	0
Example		101502100	100203200	1100503100	1201602100